customer Behavior

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```
SELECT
Product_Category,
SUM(Total_Purchase_Amount) AS most_sold_product
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1
;
```

Result Grid | Filter Rows:

Product_Category	most_sold_product
Home	171138916

- This query identifies the most sold product category by calculating the total purchase amount for each category. The data is grouped by product category, and the total purchase amount is calculated using the SUM function. The results are sorted in descending order to find the highest selling category. According to the result, the Home category has the highest total purchase amount (171138916).

```
SELECT
Payment_Method, COUNT(Transaction_id) AS
most_use_method
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

Result Grid | Filter Rows:

Payment_Method	most_use_method
Credit Card	83547

- A quantitative analysis was conducted to determine the primary payment method used by customers. By aggregating Transaction_id and grouping by Payment_Method, the result identifies Credit Card as the most frequently used payment mode, accounting for 83,547 individual transactions. This insight highlights a strong consumer reliance on credit services for their purchasing activities."

```
SELECT
gender, COUNT(customer_id) AS gender_count
FROM
customer_behaviors
GROUP BY 1
order by 2 desc;
```

Result Grid | Filter Rows

gender	gender_count
Male	125676
Female	124324

- “The customer base is slightly dominated by female customers, although the overall distribution remains fairly balanced between both genders. This indicates that while female customers present a strong opportunity for targeted marketing and higher engagement, male customers still represent a significant portion of the market. Therefore, the company should adopt a dual-segment strategy—focusing on enhancing female-oriented marketing campaigns while also developing tailored initiatives to engage and retain male customers effectively.”

```
SELECT
Product_Category,
ROUND(AVG(Product_Price), 2) AS avgerage_category
FROM
customer_behaviors
GROUP BY 1
order by 2 desc;
```

Result Grid | Filter Rows:

Product_Category	avgerage_category
Home	255.64
Books	254.72
Clothing	254.32
Electronics	254.28

- **“Since pricing across categories is very similar, the company should discuss whether to maintain a uniform pricing strategy or introduce differentiation, while also analyzing demand, profitability, and customer behavior to make more data-driven decisions.”**

```
SELECT
customer_name,
SUM(Total_Purchase_Amount) AS highest_purchase_amount
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

Result Grid | Filter Rows:

customer_name	highest_purchase_amount
Michael Johnson	339219

- “The analysis identifies the top-spending customer, highlighting a key revenue contributor. The company should focus on retaining and engaging such high-value customers through personalized and loyalty-driven strategies.”

```
SELECT
customer_name, SUM(Quantity) AS top_5_Customer_quantity
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Result Grid | Filter Rows:

customer_name	top_5_Customer_quantity
Michael Johnson	376
Michael Smith	348
Michael Brown	313
John Smith	292
Michael Jones	280

- Loyalty Rewards: Implement a VIP or loyalty program for these top 5 customers to ensure long-term retention.
- Targeted Marketing: Since "Michael" and "Smith" appear frequently, verify if these are unique accounts or if there is data duplication. If they are unique, personalized email campaigns could further increase their engagement.
- Bulk Incentives: Offer "volume-based" discounts to these individuals to encourage even larger orders in the future.
- Customer Feedback: Reach out to these specific customers for surveys or testimonials, as they have the most experience with our ordering process.

```
SELECT
customer_name, SUM(Total_Purchase_Amount) AS total_amount
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 DESC;
```

- Based on the data analysis, the company can implement the following strategic decisions:
- High-Value Segment: Offer an 'Elite Membership' to customers who have made purchases exceeding 250k. These premium customers should be provided with exclusive discounts or free delivery services to maintain their loyalty.
- Revenue Concentration: A significant number of customers share similar names and surnames (e.g., Johnson, Smith). It is crucial to verify transactions using Customer IDs to determine if revenue is heavily dependent on a few specific individuals.
- Tiered Rewards: Categorize customers into different levels (Gold, Silver, Platinum) based on their total spending. For instance, customers spending over 200k should receive a different set of perks compared to those spending below 150k.
- Retention Strategy: For customers in the lower spending bracket (140k – 160k range), the company should focus on increasing their purchase frequency. Sending targeted 'Next Purchase Bonus' coupons can encourage them to move into higher spending tiers.

customer_name	total_amount
Michael Johnson	339219
Michael Smith	321691
Michael Brown	299694
James Smith	265814
Michael Jones	265805
John Smith	258797
David Johnson	247292
John Johnson	234897
Michael Williams	221734
Jennifer Smith	220167
Robert Brown	218042
Steven Smith	203949
Jennifer Johnson	194148
Michael Taylor	193680
David Miller	189299
David Smith	181277
Matthew Smith	179962
Jennifer Jones	176943
Christopher Jones	174874
Jennifer Brown	174335
Michael Garcia	174022
Michael Thomas	173338
Christopher Jo...	172997
Kevin Johnson	165184
Joseph Johnson	164136
Michael Davis	163736
William Williams	161162
Amanda Brown	160985
Jessica Smith	160615
Angela Smith	157305
Daniel Johnson	153161
Kimberly Smith	150972
Christopher Will...	150480
Robert Johnson	149718

```
SELECT
Product_Category, COUNT(Returns) AS Returns_category
FROM
customer_behaviors
WHERE
Returns = 1
GROUP BY 1
ORDER BY 2 DESC;
```

Product_Category	Returns_category
Electronics	25448
Books	25406
Home	25320
Clothing	25302

- Based on the analysis of return counts across different product categories, the following strategic actions are recommended:
- Quality Control for Electronics: Since the Electronics category has the highest number of returns (25,448), the company should investigate potential hardware defects or shipping damages. Implementing stricter quality checks before dispatch could reduce these numbers.
- Detailed Product Descriptions for Books: The high return volume in the Books category (25,406) might suggest a gap between customer expectations and the actual product. Adding more detailed previews, "Look Inside" features, or accurate condition descriptions (for used books) may help.
- Reviewing 'Home' Category Logistics: The Home category (25,320 returns) often involves fragile or bulky items. The company should analyze if returns are due to transit damage and consider improving packaging materials to ensure items arrive safely.
- Sizing and Fit Accuracy for Clothing: For the Clothing segment (25,302 returns), the primary reason for returns is often incorrect sizing. Providing comprehensive size charts, customer height/weight reviews, or AR-based "virtual try-on" tools could significantly lower the return rate.
- Cross-Category Return Benchmarking: Interestingly, the return counts are very similar across all four categories (ranging from 25,302 to 25,448). This suggests a systemic issue in the general return policy or the fulfillment process rather than a problem with a specific product type. A full audit of the customer service and return experience is recommended.

```
select customer_name,sum(Total_Purchase_Amount) as
total_amount,
rank() over(order by sum(Total_Purchase_Amount) desc
)
as ranking from customer_behaviors group by 1;
```

Result Grid | Filter Rows:

customer_name	total_amount	ranking
Michael Johnson	339219	1
Michael Smith	321691	2
Michael Brown	299694	3
James Smith	265814	4
Michael Jones	265805	5
John Smith	258797	6
David Johnson	247292	7
John Johnson	234897	8
Michael Williams	221734	9
Jennifer Smith	220167	10
Robert Brown	218042	11
Steven Smith	203949	12
Jennifer Johnson	194148	13
Michael Taylor	193680	14
David Miller	189299	15

It is worth noting that if two customers had the exact same total_amount, the RANK() function would give them the same number and then "skip" the next number in the sequence (e.g., 1, 2, 2, 4). If you prefer a continuous sequence (1, 2, 2, 3), you would use DENSE_RANK().

```
with Purchase_customer as (
select customer_name,Product_category,Total_Purchase_Amount,
rank()
over(partition by Product_category order by Total_Purchase_Amount
desc) Purchase_ranking
from customer_behaviors
)

select * from Purchase_customer
where Purchase_ranking=1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	Product_category	Total_Purchase_Amount	Purchase_ranking
Mary Johnson	Books	5345	1
Gary White	Clothing	5350	1
Kyle Jackson	Electronics	5349	1
Joan Carson	Electronics	5349	1
Jeremiah Barnes	Home	5348	1

- Query Breakdown
- The SQL uses a Common Table Expression (CTE) named Purchase_customer to perform the following:
- Partitioning: PARTITION BY Product_category tells the database to reset the ranking every time it hits a new category (e.g., Books, Clothing, Electronics).
- Internal Ranking: Inside each category, it ranks customers from highest to lowest spender (ORDER BY Total_Purchase_Amount DESC).
- Filtering: The final WHERE Purchase_ranking = 1 filters the list to show only the top person in each category.

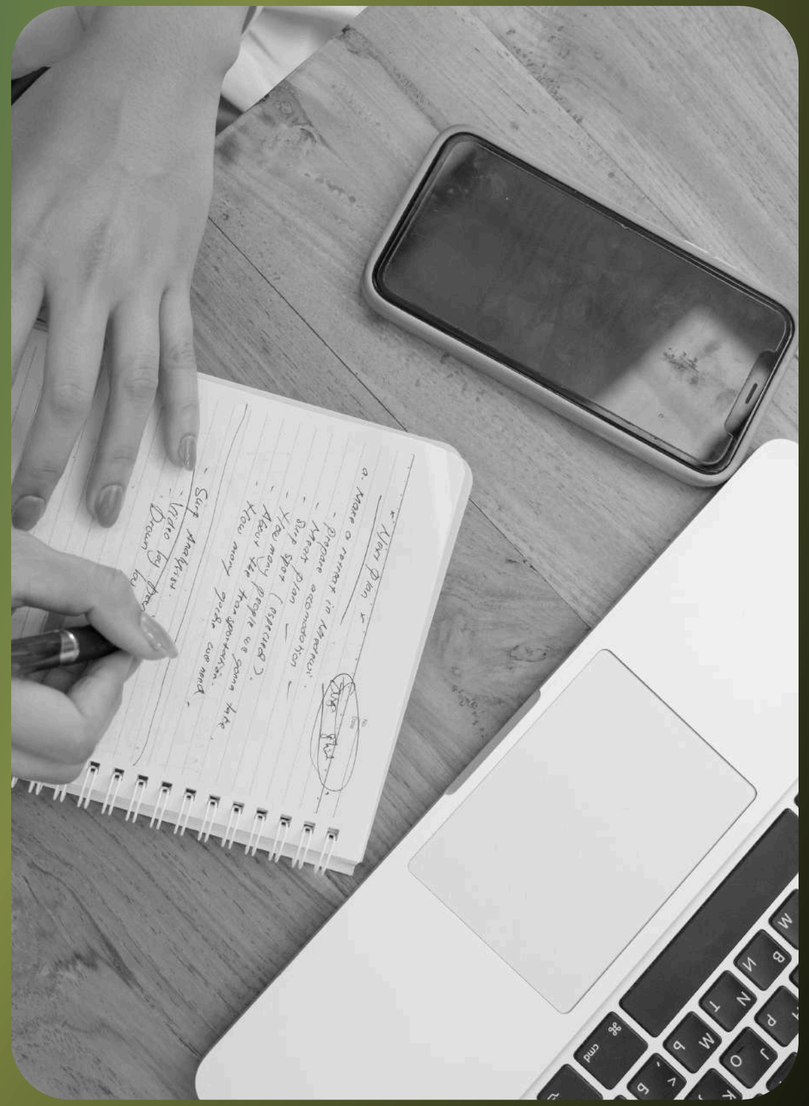
Books: Mary Johnson is the top buyer ($5,345).
Clothing: Gary White is the top buyer ($5,350).
Electronics: This is interesting! Kyle Jackson and Joan Carson are tied for 1st place with identical spending ($5,349). Because the RANK() function was used, both are assigned "1".
Home: Jeremiah Barnes is the top buyer ($5,348).

```
with top_customer as (
select customer_name,sum(Total_Purchase_Amount) as top_spending from
customer_behaviors
group by 1
)
select customer_name,top_spending from top_customer
order by 2 desc limit 5;
```

Result Grid | Filter Rows:

customer_name	top_spending
Michael Johnson	339219
Michael Smith	321691
Michael Brown	299694
James Smith	265814
Michael Jones	265805

The analysis highlights our most valuable customers based on lifetime purchase value, with Michael Johnson leading as the top contributor."



```
SELECT
SUM(Total_Purchase_Amount) AS Total_Revenue
FROM
customer_behaviors;
```

Result Grid

Total_Revenue
681346299

Total Customer Revenue

```
SELECT
Product_Category,(SUM(CASE
WHEN Returns = 1 THEN 1
ELSE 0
END) / COUNT(*)) * 100 AS Return_Rate_Percentage
FROM
customer_behaviors
group by 1
order by 2 desc;
```

Product_Category	Return_Rate_Percentage
Books	40.8100
Electronics	40.6300
Home	40.4800
Clothing	40.4300

- The data analysis reveals a critically high return rate across all product categories, averaging approximately 40.6%. This indicates a systemic issue that is likely impacting profitability due to increased reverse logistics costs and lost sales.

```
SELECT DISTINCT
customer_name,
SUM(Total_Purchase_Amount) AS Total_purchase,
AVG(Total_Purchase_Amount) AS average_purchase
FROM
customer_behaviors
GROUP BY 1
ORDER BY 2 , 3 DESC;
```

Result Grid | Filter Rows: | Export:

customer_name	Total_purchase	average_purchase
Gerald Adams	125	125.0000
Chad Torres	160	160.0000
Russell Walsh	182	182.0000
Chris Weiss	184	184.0000
Zoe Clark	194	194.0000
Nicole Foley	194	194.0000
Daniel Cohen	196	196.0000
Dr. Debra Rosario	212	212.0000
Billy Sanchez	215	215.0000
Debra Travis	220	220.0000
James Waters	247	247.0000
Jordan Hart	251	251.0000
Aaron Bennett	262	262.0000
Joe Jones DDS	265	265.0000
Erin Patel	278	278.0000
Randy Bailey	285	285.0000

Result 37

Thank You for Listening